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SUPERSEDING
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INTERIM FEDERAL SPECIFICATION

HAIRBRUSH, SYNTHETIC

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D.C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers synthetic hairbrushes (see 3.1.1 and 3.1.2), used primarily by women.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

- H-S-951 - Synthetic Filaments for Brushes, General Specifications for.
- L-P-380 - Plastic Molding Material, Methacrylate.
- L-P-394 - Plastic Molding Material (Propylene Plastics, Injection and Extrusion).
- L-P-398 - Plastic Molding Material, Styrene-Butadiene.
- L-P-410 - Plastic, Polyamide (Nylon), Rigid, Rods, Tubes, Flats, Molded and Cast Parts.
- L-P-1183 - Plastic Molding Material, Acrylonitrile-Butadiene-Styrene (ABS).
- PPP-B-566 - Boxes, Folding, Paperboard.
- PPP-B-636 - Box, Fiberboard.
- PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
- PPP-B-676 - Boxes, Set-Up.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks, as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from the Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, MO., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

- MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
- MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply

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American Society for Testing and Materials (ASTM) Standards

D 785 - Rockwell Hardness of Plastics and Electrical Insulating Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19130.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Freight Classification, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Plastic block. Unless otherwise specified, the material for plastic blocks shall be one of the following at the option of the manufacturer.

(a) Methyl methacrylate in conformance with type I, class 2 or 3 of L-P-380, except that requirements should be limited to those in table I therein.

(b) Polypropylene in conformance with type I of L-P-394 except that requirements shall be limited to density, yield strength, and impact strength. In addition, the polypropylene shall have a Rockwell hardness (R scale) of 96 to 104, inclusive, when determined as specified in ASTM D 785.

(c) Styrene-butadiene in accordance with type III, class 2 of L-P-398, except that requirements shall be limited to impact strength, tensile strength, and melting point.

(d) Nylon 6/10 in conformance with L-P-410 except that requirements shall be limited to specific gravity, Rockwell hardness (R scale), impact, strength, and melting point.

(e) Acrylonitrile-butadiene-styrene (ABS), in accordance with type I or II of L-P-1183 except that requirements shall be restricted to impact strength, Rockwell hardness (R scale), specific gravity, and tensile yield stress.

Clean unburned plastic materials in the form of imperfect parts, sprues, runners, or other scrap of the same composition as the virgin material and produced during the molding and finishing operations may be reground and mixed with the virgin material.

3.1.2 Synthetic filaments. Filaments shall be class N, type II level, in accordance with H-B-951. The ends of the filaments shall be rounded in accordance with illustrations shown in figure 1 to remove sharp edges that could cause injury to the hair and scalp. When examined as specified in 4.3.1, not more than 25 percent of the filaments shall be classified as defective.

3.1.2.1 Diameter of filaments. The average diameter of the filaments shall be 0.022 inch within the tolerances specified in H-B-951 (see 4.3.2).

3.1.3 Staple wire. Staple wire shall be inherently corrosion resisting steel wire or low carbon steel wire coated with zinc or other suitable metal to resist corrosion under normal service conditions. The diameter of the wire shall be not less than 0.181 (U. S. Steel Wire Gage No. 26).

3.1.4 Design. Brushes shall be designed to be held conveniently in the hand while brushing the hair and to brush the hair without injury to the hair or scalp. The shape of the brush shall be the manufacturer's option within the limitations imposed by this specification.

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3.3 Construction.

3.3.1 Block (including handle). The block shall be molded in one solid piece. The surface shall be free from sharp edges. The dimensions of the block shall be as specified in table I. The face of the block at the brush end shall have not less than 72 holes for tufts. The holes shall be distributed over the block in not less than five staggered rows.

TABLE I. Dimensions of block

Dimensions	Required values
	inches
Length of block, overall (including handle) minimum	7-7/8
Thickness of block (including handle and neck portion) minimum	5/16
Face of block (area for distribution of holes).	
Length (edge-to-edge):	
At center, minimum	4
At sides, minimum	3-5/8
Width at any point, minimum	1-1/8
Width of neck portion, minimum	7/16

3.3.2 Tufts.

3.3 2 1 Length of filaments out of block. When determined as specified in 4.3 3, the filaments in each tuft shall range in length out of the block from 1/2 to 1-1/8 inch, inclusive, but not more than 2 percent of the filaments shall measure less than 7/8 inch and no filament shall measure less than 1/2 inch.

3 3 2.2 Filament count per tuft. When determined as specified in 4 3.4, the average number of filament ends per tuft shall be not less than 12 (6 filaments), and no tuft shall have less than 10 filament ends (5 filaments)

3 3 2 3 Flare of tufts. All horizontal rows of tufts shall flare gradually from the center toward opposite ends attaining an angle of 10 ± 5 degrees at each end. Likewise, all lengthwise rows, except the center row, shall flare gradually from the center row in opposite directions toward each side of the block, attaining an angle of 25 ± 5 degrees at each side. The angle of flare shall be determined as specified in 4.3.5.

3.3.2 4 Set of tufts. The tufts of filaments in each brush shall solidly fill the holes in the block (see 3.3.1). A staple (see 3.1 3) spanning the filaments in each tuft midway between the ends and driven into the bottom of each hole shall secure the filaments in the block so that they cannot be dislodged when tested as specified in 4.3 6. There shall be no distortion or cracks in the block and no deformed filaments due to the stapling operation.

3.4 Resistance of block to impact. When dropped from a height of 6 feet onto a hard surface such as marble, concrete, or stone, as specified in 4.3.7, brush blocks shall not break, chip, or develop defects that affect their serviceability.

3.5 Weight of brush. Each brush shall weigh not less than 2-1/16 ounces (see 4.3.7).

3.6 Identification of class N filaments. In case of dispute, nylon filaments in the finished brush shall meet the following requirements.

3.6 1 Burning characteristics. When ignited as specified in 4.3.9 1, nylon filaments from the finished brush shall produce a blue flame with yellow tip, melt, drip, and have the aroma of burned wool.

3 6 2 Specific gravity. When tested in monochlorobenzene (specific gravity 1.10) as specified in 4.3.9.2, nylon filaments from the finished brush shall float in the liquid medium.

3.6.3 Resistance to concentrated hydrochloric acid. When tested in concentrated hydrochloric acid as specified in 4 3.9.3, nylon filaments from the finished brush shall not completely dissolve.

3.7 Color.

3.7.1 Blocks Unless otherwise specified, the color of brush blocks shall be the manufacturer's standard color or assorted standard colors (see 6.1). The assorted standard colors when supplied shall be in multiples of not less than 12 of each color furnished. (See 6.1.)

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3.7 2 Filaments. Unless otherwise specified, the filaments in the brush part shall be neutral color (See 6.1.)

3 8 Marking for idertification Unless otherwise specified, brushes shall be permanently and legibly marked with the name, or trademark easily identifiable with the manufacturer, or both

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements

4.1 1 Certificates of compliance Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certificates

4 2 Sampling for inspection (examination and testing) and acceptance. Sampling for examination, testing, and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be examined and tested in accordance with all the requirements of referenced specifications drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 End item examination and testing.

4.2.2.1 Lot. For the purpose of examination and testing of finished brushes, a lot shall, as far as is practicable, consist of all brushes made of the same materials manufactured under essentially the same conditions and at essentially the same time.

4.2.2.2 Visual examination. Examination shall be made of the end item for defects listed in table II. the Inspection Level shall be II and the Acceptable Quality Level (AQL) 4.0 defects per 100 units The sample unit shall be one brush. If the number of defects in the sample for examination exceeds the acceptance number specified in MIL-STD-105, the entire lot of brushes shall be considered as having failed to meet the requirements of the specification.

TABLE II. Visual defects

Examine	Defects
Construction:	
Block	Sharp corners or edges.
	Rough, pitted, or cracked surfaces.
	Color not as required (when specified).
	Misshapen.
	Dirty or embedded with foreign material.
Tufts:	Deformed filaments.
	Color not as required.
	Tufts do not solidly fill holes.
	Less than minimum number of tuft holes.
	Filaments soiled or embedded with foreign materials.
	Not firmly anchored - can be easily removed by hand.
Workmanship	Defects present which affect appearance or serviceability.
Marking:	Omitted.
	Illegible.

4.2.2.3 Dimensional examination. Examination shall be made of the end item for compliance with the dimensional requirements of table I. The Inspection Level shall be S-2 and the Acceptable Quality Level (AQL) 4.0 percent defective. The sample unit shall be one brush. If the sample unit fails to meet any dimensional requirement specified in table I, it shall be considered defective. If the number of defective sample units in the sample for test exceeds the acceptance number specified in MIL-STD-105, the entire lot of brushes shall be considered as having failed to meet the requirements of the specification

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4 2 2 4 Testing The end item shall be tested as specified in 4 3 through 4 3 9.3 The Inspection Level shall be S-1 and the Acceptable Quality Level (AQL) 4 0 percent defective The sample unit shall be one brush If the sample unit fails to meet any requirement for which tests are made, it shall be considered defective If the number of defective sample units in the sample exceeds the acceptance number in MIL-STD-105, the entire lot of brushes shall be considered as having failed to meet the requirements of the specification

4 2 2 5 Examination of preparation for delivery An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5 of this specification Defects shall be scored as listed in table III Sampling shall be in accordance with MIL-STD-105 The sample unit shall be one shipping container fully prepared for delivery The lot shall be the number of containers offered for inspection at one time The Inspection Level shall be S-2 with an Acceptable Quality Level (AQL) of 4 0, expressed in terms of defects per hundred units

TABLE III Examination of preparation for delivery

Examine	Defect
Containers	Not as specified
Contents	Not as specified
Markings	Omitted, incorrect, illegible, improper size, location, sequence or method of application.
Materials	Component missing or damaged. Bulging or distortion of containers.
Workmanship	Cushioning inadequate, improper or missing.

4.2.2.5.1 Examination for closure, waterproofing, and reinforcement of containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing and reinforcement shall be in accordance with the appendix of that specification.

4.3 Tests for end item.

4.3.1 Microscopic examination of filament ends. Examine six tufts located at representative places on the block with a 10-power (10X) lens and a setting of 1.5 (15X magnification) for rounding of filament ends. Record the number of unacceptable filaments in the six tufts as A. Count the total number of filament ends in the six tufts and record the value as B. Calculate the percentage of defective filaments as follows and check the result for conformance with 3.1.2:

$$\text{Percentage of defective filaments} = \frac{A}{B} \times 100$$

4.3.2 Diameter of filaments. The average diameter of the filaments in the finished brush shall be determined by the applicable method in H-S-951 and the result checked for conformance with 3.1.2.1. The specimen shall be composed of filaments from tufts located at representative places on the block.

4.3.3 Length of filaments. Measure the overall height of 15 tufts located at representative places on the block and record each value. Check each value for conformance with the applicable requirement in 3.3.2.1. Examine each of the fifteen tufts for filaments measuring less than 1/2 inch out of the block for conformance with the applicable requirement in 3.3.2.1. Determine the number of filaments measuring less than 7/8 inch and 1/2 inch or more in length clear of block. Record the result as A and check for conformance with the applicable requirement in 3.3.2.1. Count the total number of filaments in the fifteen tufts and record the result as B. Calculate the percentage of defective filaments (less than 7/8 inch and 1/2 inch or more in length clear of block) and check the result for conformance with 3.3.2.1:

$$\text{Percentage of defective filaments} = \frac{A}{B} \times 100$$

4.3.4 Filament count per tuft. Count the filament ends in fifteen tufts located at representative places on the block. Record the count for each tuft and check each value for conformance with the applicable requirement in 3.3.2.2. Calculate the average value for the fifteen tufts and check the result for conformance with the applicable requirement in 3.3.2.2.

4 3.5 Row and tuft angle. Place the brush on a level surface with the block resting on the surface. By means of a bevel protractor determine the gradual flare and angularity of the tufts at the end and side rows and check the results for conformance with 3.3.2.3.

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4.3.6 Tuft retention. Not less than five tufts from each brush shall be tested, one from each side row, one from each end row, and one from the center of each remaining lengthwise row as follows: Secure the brush in a suitable clamp and the filament ends of the tuft in another clamping device so that the filaments are evenly distributed and so as to prevent distortion and cutting. By means of a spring scale or weight, apply a pulling force of 8 ± 0.5 pounds in a direction perpendicular to the block to determine conformance with 3.3.2.4.

4.3.7 Resistance of plastic block to impact. Drop the brush five times from a height of 6 feet onto a hard surface such as marble or concrete so that the block strikes the surface each time. After each drop, examine the block for conformance with 3.4.

4.3.8 Weight of brush. Weigh the brush on a suitable scale and check the result for conformance with 3.5.

4.3.9 Identification of nylon filaments in er item. Ir case of dispute regarding the identity of the filaments in the brush, filaments shall be tested as follows:

4.3.9.1 Burning test. Remove five tufts at random from the brush. Take one filament from each tuft and reserve the remaining filaments for the tests in 4.3.9.2 and 4.3.9.3. Test each filament as follows: Grasp the filament at one end with forceps or pliers and bring the free end of the filament into the flame provided by a cigarette lighter, match, or gas-burner. Hold the filament in the flame until it ignites. Care should be taken to prevent the molten plastic from coming into contact with the skin and to avoid deep inhalation of the fumes. Observe the color of the flame and the aroma given off by the burning filament for conformance with 3.6.1.

4.3.9.2 Specific gravity. Take one filament from eacn tuft in 4.4.9.1. Drop the five filaments into a suitable glass container holding sufficient monochlorobenzene (specific gravity 1 10) to permit the filaments to circulate freely in the solution. With a suitable instrument, poke the filaments beneath the surface of the solvent to thoroughly wet them. Observe each filament for conformance with 3.6.2.

4.3.9.3 Resistance to concentrated hydrochloric acid. Place five filaments one from each tuft in 4.4.9.1 in a suitable glass container. Cover the filaments with concentrated hydrochloric acid (37 - 38 percent, specific gravity 1.19), and allow them to remain immersed in the acid for one hour. Care should be taken to prevent the acid from coming into contact with the skin or clothing. Examine the filaments for conformance with 3.6.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Level A, B, or C packaging shall be as specified (see 5.1.1 and 5.1.1.1).

5.1.1 Unit packaging. Each hairbrush shall be packaged in a container commonly used for the item and which affords adequate protection against damage during shipment from the supply source to the first receiving agency.

5.1.1.1 Intermediate packaging. Twelve (12) unit packages as specified in 5.1 shall be intermediately packaged in a snug-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. The brushes in quantities as specified (see 6.1), shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed, water-proofed, and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. Twelve (12) intermediate packages of brushes shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed, water-proofed, and strapped in accordance with the appendix to the box specification.

5.2.3 Level C. Hairbrushes in quantities as specified (see 6.1), packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special markings required in the procurement documents, interior packages and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123 for civil agencies or MIL-STD-129 for military agencies, as applicable.

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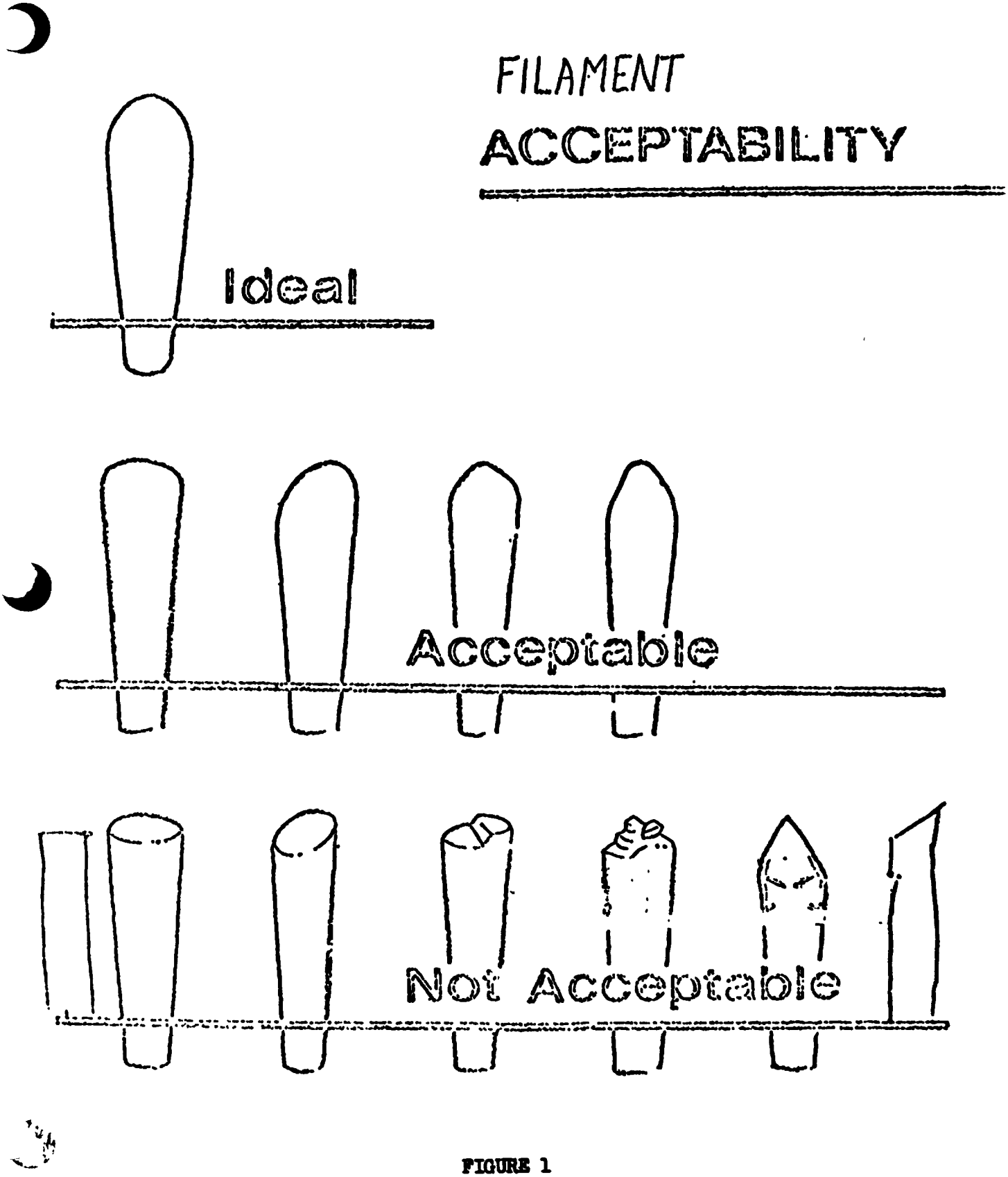


FIGURE 1

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6. NOTES

6.1 Ordering data. Purchasers should exercise any options offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a specific plastic material is required for the block (see 3.1.1).
(c) When a specific color is required for the block (see 3.7.1).
(d) When marking as specified in 3.8 is not required.
(e) Selection of applicable levels of packaging and packing (see 5.1 or 5.2).
(f) Quantities required (see 5.2.3).
(g) Marking document required (see 5.3).

Civil Agency Coordinating Activities:

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